

Introduction

XML and its related technologies form a crucial part of modern web development. They enable structured data storage, document interchange, and cross-platform communication. While HTML defines how data is displayed, XML defines how data is **structured** and **shared** across systems. XHTML and XHTML-MP are specialized applications derived from XML, designed for web and mobile platforms respectively.

1. XML (eXtensible Markup Language)

Definition

XML (eXtensible Markup Language) is a markup language designed to store and transport data in a structured, platform-independent format. It was developed by the **World Wide Web Consortium (W3C)** in **1998** as a simplified subset of SGML.

Unlike HTML, XML does not describe how data looks but focuses on what the data **is**. It provides a framework for defining custom tags and allows applications to read and process data consistently across different environments.

Features of XML

1. **Extensible:** Users can create their own tags based on their data needs.
 2. **Self-Descriptive:** Each piece of data is enclosed in tags that explain its meaning.
 3. **Platform and Language Independent:** Works across all systems and programming languages.
 4. **Structured and Hierarchical:** Organizes data in a tree-like structure, allowing easy parsing and modification.
 5. **Separation of Data and Presentation:** Focuses on content rather than display.
 6. **Supports Validation:** XML documents can be validated using DTD or XML Schema.
 7. **Unicode Support:** Handles multiple languages and character sets.
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Basic Syntax of XML

An XML document consists of elements, attributes, and proper nesting.

Example:

```
<?xml version="1.0" encoding="UTF-8"?>
<student>
  <name>Noor Fatima</name>
  <rollno>BSCS-2025</rollno>
  <department>Computer Science</department>
</student>
```

Explanation: - The declaration `<?xml version="1.0" encoding="UTF-8"?>` defines the XML version and character encoding.

- `<student>` is the **root element**.
- Child elements `<name>`, `<rollno>`, and `<department>` store specific data values.
- XML is case-sensitive and requires proper nesting and closing of all tags.

XML Rules

1. Every XML document must have **one root element**.
2. All elements must have **closing tags**.
3. XML is **case-sensitive**.
4. Attributes must be enclosed in quotes.
5. Tags cannot overlap.
6. Comments are written as `<!-- comment -->`.

Well-Formed vs Valid XML

Type	Description	Example
Well-Formed XML	Follows XML syntax rules (proper nesting, closing tags).	<code><book><title>PUACP Guide</title></book></code>
Valid XML	In addition to being well-formed, it adheres to a DTD or XML Schema.	<code><!DOCTYPE book SYSTEM "book.dtd"></code>

DTD (Document Type Definition)

A **DTD** defines the structure and rules of an XML document. It specifies what elements and attributes are allowed.

Example:

```
<!DOCTYPE student [
  <!ELEMENT student (name, rollno, department)>
```

```
<!ELEMENT name (#PCDATA)>
<!ELEMENT rollno (#PCDATA)>
<!ELEMENT department (#PCDATA)>
]>
```

This ensures all XML documents follow a specific format.

Advantages of XML

- Facilitates data sharing between different systems.
- Platform and language independent.
- Easily readable by both humans and machines.
- Useful for configuration files and web services.
- Supports hierarchical data.

Disadvantages of XML

- Verbose and large file sizes.
 - Complex to parse for beginners.
 - Slower to process compared to lightweight formats like JSON.
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2. XHTML (Extensible HyperText Markup Language)

Definition

XHTML is a hybrid between **HTML** and **XML**. It combines the presentation structure of HTML with the strict syntax rules of XML. It was developed by W3C to make HTML documents more structured, consistent, and compatible with XML tools.

XHTML documents must be **well-formed** and **valid**, ensuring compatibility with modern browsers and XML parsers.

Features of XHTML

1. Based on XML syntax but displays like HTML.
 2. Every tag must be closed properly.
 3. All element and attribute names must be in lowercase.
 4. Attribute values must be quoted.
 5. Proper nesting of elements is required.
 6. Allows integration with other XML-based languages (like SVG, MathML).
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Example of XHTML

```
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Strict//EN" "http://www.w3.org/TR/
xhtml1/DTD/xhtml1-strict.dtd">
<html xmlns="http://www.w3.org/1999/xhtml">
<head>
  <title>XHTML Example</title>
</head>
<body>
  <h1>Welcome to PUACP</h1>
  <p>This is an example of a well-formed XHTML document.</p>
  
</body>
</html>
```

Explanation: - The `xmlns` attribute defines the XML namespace.

- Every tag is closed (e.g., `<img />`).

- Lowercase elements and attributes ensure XML compliance.

Advantages of XHTML

- Ensures clean, well-structured code.
- Compatible with both browsers and XML tools.
- Easier to maintain and validate.
- Improves accessibility and cross-device support.

Disadvantages

- Stricter syntax than HTML.
- Minor syntax errors can break the entire page.
- Requires developers to strictly follow XML rules.

3. XHTML-MP (XHTML Mobile Profile)

Definition

XHTML-MP (XHTML Mobile Profile) is a lightweight version of XHTML designed specifically for **mobile devices**. It was developed by the **Open Mobile Alliance (OMA)** and became the standard for early mobile browsers before responsive web design.

Features of XHTML-MP

1. Simplified syntax and smaller file sizes for faster loading.

2. Optimized for limited memory and bandwidth environments.
3. Uses a subset of XHTML and CSS for mobile display.
4. Supports forms, links, and images suitable for small screens.
5. Compatible with WAP 2.0 (Wireless Application Protocol).

Example of XHTML-MP

```
<?xml version="1.0"?>
<!DOCTYPE html PUBLIC "-//WAPFORUM//DTD XHTML Mobile 1.2//EN" "http://
www.openmobilealliance.org/tech/DTD/xhtml-mobile12.dtd">
<html xmlns="http://www.w3.org/1999/xhtml">
<head>
  <title>PUACP Mobile</title>
</head>
<body>
  <h2>Welcome to PUACP Mobile Portal</h2>
  <p>This lightweight page is optimized for mobile browsing.</p>
</body>
</html>
```

Advantages of XHTML-MP

- Efficient performance on mobile devices.
- Standardized mobile markup language.
- Reduces bandwidth and data load.
- Compatible with mobile browsers and WAP 2.0.

Disadvantages

- Limited multimedia and styling support.
- Outdated for modern smartphones.
- Replaced by responsive HTML5 and CSS3 frameworks.

Summary Table

Language	Definition	Purpose	Example
XML	Extensible Markup Language	Defines and structures data	<code><student><name>Noor</name></student></code>
XHTML	Extensible Hypertext Markup Language	Combines HTML and XML syntax	<code><p>Hello</p></code> (well-formed)

Language	Definition	Purpose	Example
XHTML-MP	XHTML Mobile Profile	Mobile-friendly version of XHTML	<code><h2>Mobile Page</h2></code>

Conclusion

XML and its derivatives — XHTML and XHTML-MP — provide a standardized way to represent, exchange, and display structured data across different platforms. While HTML5 dominates the modern web, XML remains essential in web services, data storage, and backend communication systems.

